GaneshKumar Karuppaiah

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PROFESSIONAL SUMMARY

Successful Software Engineering Professional with 13+ years of experience in designing, developing, and managing distributed systems, enterprise applications, and scalable full-stack solutions. Strong expertise in Java, Spring Boot, microservices, Python, and full-stack development, combined with specialization in Large Language Models (LLMs), Generative AI, and cloud-based AI pipelines across AWS, Azure, and GCP. Proven expertise in the end-to-end software and ML lifecycle, including application architecture, API development, data ingestion, model fine-tuning, Retrieval-Augmented Generation (RAG), evaluation, and deployment of production-ready solutions. Experienced in building enterprise-grade GenAI applications using LangChain, Hugging Face, TensorFlow, and PyTorch. Adept at technical leadership, mentoring engineers, designing robust solutions, and delivering secure, scalable, and high-performance applications aligned with business requirements.

EDUCATION

M. Tech. in AIML - Bits Pilani	81%	2025
B. E in ECE - Anna University	73%	2013
HSSC	87%	2009
SSLC	76.6%	2007

WORK HISTORY

Bank of Julius Baer – Senior Technical Lead | Full stack engineer
One Changi City, Singapore - 01/2026 – current
Project 1: Global Output Management

Bank of New York – Lead AI Full Stack Developer
Chennai, India 08/2020 – 01/2026
Project 1: Newton Performance product – Aladdin
Project 2: Newton Client Reporting – Aladdin
Project 3: Newton Portfolio Transaction Management – Aladdin

Cognizant Technologies Solutions – Lead Full Stack Developer
Chennai, India 07/2016 – 08/2020
Project 1 – Verizon Broadband Gateway (BGW) UUI 2.0
Project 2 – Verizon XO Migration (BGW) UUI 3.0

Oviya Technologies Pvt Ltd - Software Developer

Bangalore, India. 06/2013 – 07/2016

Project: Car Digital Video Recording – (DVR, SWED, MERDS)

Roles and Responsibilities

- ✓ Expertise in Backend & frontend programming languages
- ✓ Proficiency in AI and machine learning algorithms
- ✓ Experience with AI frameworks, did Fine-tuning RAG, GPT LLMs
- ✓ Enhanced code quality and scalability through rigorous code reviews, repository management, and CI/CD optimization.
- ✓ Successfully led cross-functional teams to deliver innovative AI solutions for enterprise and consumer-facing products.
- ✓ Understanding of deep learning architectures such as CNNs, RNNs, and LSTMs
- ✓ Ability to design and implement AI-based solutions
- ✓ Involved in Requirement Analysis, Development, Testing & Architecture level designing.
- ✓ Analyzed existing functionalities and developed the UI & Rest Web Services for the application.
- ✓ Integrated the Rest APIs with User Interface.

SKILLS & ENVIRONMENTS

Java 25, Spring Boot, Microservices, Rest, Oracle, Snowflake, Snowpipe, Cortex AI, Langchain, Liquibase, Angular 20, ES6, Kafka, camel, Splunk, Python, TensorFlow, PyTorch, Keras, Scikit-learn, Transformers, Hugging Face, SpaCy, NLTK, FAISS OpenCV, PySpark, NLP, RAG, Jupyter,Pytest. OOPS, Tableau, App dynamics, app engine, Grafana, Rust, RdKafka, Rest API, Oracle, Cron, Sonar-Rust, cucumber, Jest, SonarQube, Jira, confluence, Azure.

J2EE, Spring, Hibernate, REST, Beans, Servlets, Swing, JSP, Applets, C, C++, Typescript, Java Script, HTML5, CSS, Less, JQuery, Ajax, JSON, Properties, Maven, Gradle, XML, Angular Bootstrap UI, Bean shell, React-JS, Jest, Jasmine, Karma, Cordova, Phone-gap, Node JS, Android, Log4j.

AWS, My SQL, SQLite, Oracle SQL, Apache Tomcat, GitLab, JIRA, Jenkins, Putty, Microservices, Stash, Confluence, Github, Docker.

Android Studio, Net Beans, Eclipse, XAMPP, Qt–Embedded, Postman, Agile Methodology, Kanban (scrum). Experienced with AWS, GCP, Azure, and integration with Hugging Face and GitHub. Proven ability to lead teams, mentor engineers, and foster software best practices.

Windows, Ubuntu, Debian, Linux

CERTIFICATIONS

Oracle Database 10g& 11g Administrator Certified Professional (OCA, OCP).

Oracle Certified Java Professional (OCJP)

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- ✓ Designed Mesh Network Algorithm in application.
- ✓ Implemented & Tested JUnit, Integration, Regression Tests.
- ✓ Played major role in team for story deliveries.
- ✓ Coordinated and code reviewed for the team members, bug fixes are resolved on time.
- ✓ Onshore user interaction for major changes implementation with application changes.
- ✓ Proven ability to lead teams, mentor engineers, foster software best practices and delivery excellence, currently leading a **6-member team** and previously managed a **16-member engineering team** delivering scalable enterprise applications.

Amazon Web Services - Developer Associate (AWS)

Microsoft Azure Solutions Architect Expert
Snowflake – SNOWPRO Associate

Claude with Google vertex AI



Project Summary

Skills Utilized

Bank Of Julius Baer – Senior Technical Lead | Full stack Engineer — **Duration: 01/2026 to present**

Project 1 – Global Output Management - Zurich

• Developed an event-driven Global Output Management platform for processing transactions, statements, trade confirmations, corporate actions, tax documents, and client communications across Equities, Securities, FX, Derivatives, and Precious Metals.
• Designed and developed data integration pipelines to source banking data from Temenos Transact (T24) and TAP, apply business rules and transformations, and publish data to downstream systems for reporting and business operations.
• Implemented event consumption and validation using Apache Kafka and IBM MQ, leveraging Java 25, Spring Boot 3, microservices, Avro, and Apache Flink (Ververica). Integrated Quadient Scaler for document generation and AWS OpsDocs for document storage and archival.
• Integrated Snowflake, Snowpipe and Snowflake Cortex AI for data ingestion, analytics and AI-assisted insights, along with Claude AI/LLMs for intelligent document processing. Built and deployed services using Docker, Kubernetes, React 19, Oracle, Flyway and AWA, with downstream publishing and client delivery through Email, Hotmail and e-post, notification channels. Implemented an AI Output Failure Resolution Agent with Agentic AI operations assistant using Claude AI and Snowflake Cortex to monitor output-processing exceptions, analyze root causes, validate business conditions, recommend/trigger remediation workflows and notify operations teams.

Java 25, Spring Boot 3, Microservices, Kafka, IBM MQ, Avro, Snowflake, Snowpipe, Snowflake Cortex AI, Claude AI, Agentic AI, Agents Skills, Harness, RAG, LLM, AWS, Amazon Bedrock, Titan, AIops, LLMOps, LangChain, Kubernetes, Docker, Apache Flink, Ververica, Oracle, Flyway, React 19, Quadient, Scaler, AWA, REST APIs, Event-Driven Architecture. Python 3.13, Zafire, MCP, Thymeleaf, Guardrails, Prompt Engineering, Security control.

Bank of New York – Lead AI Full Stack Developer — **Duration: 08/2020 to 01/2026**

Project 1 – Newton Performance Product – Aladdin

Generating the investment performance returns from the portfolio and accounting information. Performance applications will consume all the portfolio accounting data for seg and pooled clients and publish daily, monthly, attribution calculation returns.
Architected a predictive analytics tool for predicting market trends, achieving an 80% accuracy rate, thereby informing in strategic investments.
Architected predictive analytics tools and LLM-based AI chatbots are integrated with enterprise data pipelines.

Project 2 – Newton Client Reporting – Aladdin

Architected a NPF chatbot using NLP, improving client service response times by 60%. Managed data preprocessing strategies, reducing data redundancy. Built successful portfolio predictive analytics models, to enhanced to make business decision. Improved

Java 21, Oracle, Angular 14, Spring Boot, microservices, Snowflake, Snowpipe, Cortex AI, AWS, Jest, Docker, Kafka, Camel, Jest, Cucumber, Test Cafe, SonarQube, SQL, Gitlab, App engine, App dynamics, Grafana, Veracode scan, Jira, Confluence, Docker, Giffy, copilot.

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data collection methods, increasing data accuracy. Implemented clustering algorithms, improving customer segmentation efforts. Enhanced an existing recommendation system, increasing user interaction rates by 20%, Implemented reinforcement learning algorithms, achieving a 95% success rate in dynamic environments.
Developed NLP-driven Client Reporting System using LangChain and FAISS RAG architecture.

Project 3 – Newton Portfolio Transaction Management – Aladdin

A Rust-based solution for managing and analyzing financial portfolios. It processes transaction data, generates financial reports, and exposes REST APIs for integration. Features include Kafka for real-time streaming, scheduling for periodic operations, and comprehensive testing for reliability and scalability. Expertise in Kafka integration, REST API development, and scheduler implementation. Designed SQL schemas, created financial reports. Implemented CI/CD pipelines, conducted code reviews, and collaborated with data engineers and business analysts.

Awards: BNY Hackcelerator 2.0 AI Innovator Award – Portfolio-IQ chatbot (2024)

Python, TensorFlow, PyTorch, Keras, Scikit-learn, NLTK, Spacy OpenCV, PySpark, Jupyter, Airflow, MLFlow, MLops, FAISS, LangChain, AutoGen, CrewAI Nuggets, DataIQ, Databricks. FastAPI, Flask, Django. Rust, Kafka, Rest API, Oracle Cron Schedulers, Tokio, Serde, Serde-Json, Cargo, pdf-create, excel-rs, Gitlab, SonarQube, Soar-rust, cargo-tarpaulin, Python, numpy, pandas, tensorflow NLP, ADAM classifier, LSTM, AIML

Cognizant Technologies Solutions - Lead Full Stack Developer — **Duration 07/2016 to 08/2020**

Project 1 – Verizon XO Migration (BGW) UUI 3.0
Duration: 08/2019 to present.

The Telecom Switching Equipment's, cables connecting them, patch panels that are wired to equipment's including the sites/Locations where they are physically present. Entire logical inventory data are migrated to BGW. Verizon sells capacity to Enterprises.
Verizon Rapid Delivery interface for creating third party order to process XO workflow. Migrated logical inventory systems to cloud-native microservice architecture.

Java 8, React JS, Angular 6+, Spring Boot, AWS, Jest, Docker, Kubernetes, Jasmine, Karma, microservices, SQL, Gitlab, Jenkins, Agile.

Project 2: Verizon Broadband Gateway (BGW) UUI 2.0 — **Duration: 07/2016 to 08/2019**

Broadband Gateway (BGW) - Developed Ring and Mesh algorithms for Broadband Gateway (BGW), an OSS for Verizon Business, managing Layer1 inventory and provisioning, including topology building and capacity activation for customer circuits.

Core Java, Java script, JQuery, HTML5, CSS3, Spring MVC, SQL, Stash, SonarQube, JDBC, Jenkins, Jira, Swagger, windows.

Oviya Technologies Pvt Ltd – Software Developer — **Duration 06/2013 to 07/2016**

Projects: Car Digital Video Recording – (DVR, SWED, MERDS)

Implemented live vehicle tracking and video streaming, storing location data via REST APIs, processing with ActiveMQ, and displaying real-time tracking and video on a map. Tech Skills: Apache Tomcat 8, Android Studio, XML, Smart watch, Gradle, Beans, Swing, JSP, Xml, CSS, Qt-4.3 & Ubuntu Linux.

Java, J2ee, spring MVC, hibernate, html5, angular JS, Angular UI, rest web services, Net beans IDE 8.0.2, MySQL -7, SQLite, Oracle, CPP, C.

Oviya Technologies Pvt Ltd – Software Developer — **Duration 06/2013 to 07/2016**

Projects: Car Digital Video Recording – (DVR, SWED, MERDS)

Implemented live vehicle tracking and video streaming, storing location data via REST APIs, processing with ActiveMQ, and displaying real-time tracking and video on a map. Tech Skills: Apache Tomcat 8, Android Studio, XML, Smart watch, Gradle, Beans, Swing, JSP, Xml, CSS, Qt-4.3 & Ubuntu Linux.

Java, J2ee, spring MVC, hibernate, html5, angular JS, Angular UI, rest web services, Net beans IDE 8.0.2, MySQL -7, SQLite, Oracle, CPP, C.